Taxation on Savings and Capital Income
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ECO 354: Public Economics
Subhra Bhattacharya

Shiv Nadar University

Motivation

1. Capital income is about 25-30% of national income (labor income is 70-75%) but distribution of capital income is much more unequal than labor income
   Capital income inequality is due to differences in savings behavior but also inheritances received
   $\Rightarrow$ Equity suggests it should be taxed more than labor income.
2. Capital Accumulation correlates strongly with growth [although causality link is not obvious] and capital accumulation might be sensitive to the net-of-tax return.
   $\Rightarrow$ Efficiency cost of capital taxation might be high.

Motivation continued...

1. Capital more mobile internationally than labor
   Key distinction is **residence** vs. **source** base capital taxation:
   Residence: Tax based on residence of owner of capital.
   Most individual income tax systems are residence based (with credits for taxes paid abroad)
   Source: Tax based on location of capital
   Real estate property tax and corporate income tax are source based
2. Capital income taxation is extremely complex and provides many tax avoidance opportunities particularly for multinational firms (Gravelle 1994)

Wealth: definitions and facts

Definition: Capital Income = Return from Wealth
Wealth arises from expected future income and value of assets
Private wealth includes real estate (land+buildings), corporate and business equity, fixed claimed assets (bonds+deposits), net of debts (mortgage, student loans, consumer credit)
Aggregate US **Private** Wealth $\simeq 5\times$Annual National Income
Total wealth reflects both capital stock accumulated through savings and pure price effects
Example 1: house can increase in value because it is improved (capital) or because local prices go up pure price effect
Example 2: greater monopoly power makes a business more valuable to owners (but at the expense of consumers)
Recent increase in US private wealth mostly due to price effects

Wealth and capital income: facts

Wealth $= W$, Return $= r$, Capital Income $= rW$

$$W_t = W_{t-1} + r_t W_{t-1} + E_t + I_t - C_t$$

where W_t is wealth at age t, C_t is consumption, E_t labor income earnings (net of taxes), r_t is the average (net) rate of return on investments and I_t net inheritances (gifts received and bequests - gifts given).

Differences in Wealth and Capital income due to:

1) Age
2) past earnings, and past saving behavior $E_t - C_t$ [life cycle wealth]
3) Net Inheritances received I_t [transfer wealth]
4) Rates of return r_t

Wealth inequality (Saez and Zucman, 2016)

1. Wealth inequality is very large (always much higher than income inequality)
2. In the US in 2016: Top 1% wealthiest families get 40% of total wealth, Next 9% get about 35%, next 40% get 20%, bottom 50% get about 0%
   1. Wealth inequality decreases from 1929 to 1980: wealth democratization due to rise in home-ownership and pensions.
   2. Wealth inequality increases sharply since 1980 fueled by increases in **income** inequality and **savings** inequality [bottom 90% saves zero in net since 1990]
3. US public underestimates extent of wealth inequality and thinks the ideal wealth distribution should be a lot more equal [Norton-Ariely '11]

Facts on US capital income taxation

1) **Corporate Income Tax** (fed+state) on profits of corporations [complex rules with many industry specific provisions]: effective tax rate only 16% of corporate profits in 2018

2) **Individual Income Tax** (fed+state): taxes many forms of capital income

Realized capital gains and dividends receive preferential treatment (to lower double taxation of corporate profits)

Imputed rent of home owners and returns on pension funds are exempt

3) **Estate tax:** tax on very large estates (40% tax above $11m) bequeathed to heirs (small and poorly enforced)

4) **Property taxes** (local) on real estate (old tax):

Tax varies across jurisdictions. About 0.5% of market value on average

5) **Wealth tax** on total net worth of rich families (does not currently exist, proposed by Warren and Sanders)

Life cycle vs Inherited wealth

Economists divide existing wealth into 2 categories:
1) Life-cycle wealth is wealth from savings earlier in your life
2) Inherited wealth is wealth from inheritances received
Distinction matters for taxation because individuals are responsible for life-cycle wealth but not inherited wealth
Inherited wealth used to be very large in Europe (before World-War I), became small in post-World War II period, but is growing in recent decades (especially in Europe) [Piketty' 14]
Same trend in the US but less pronounced but poor data quality (Alvaredo-Piketty-Garbinti '17)
Piketty '14: return on wealth bigger than growth rate ($r > g$) $\Rightarrow$ wealth concentration and inherited wealth increases

Life cycle model

1. Overlapping generations of population, where an individual lives for two periods: Period 1 and 2.
   1. Period 1: Individual supplies l units of labour, earns ωl amount of income, consumes c_1, saves s, and retires. Savings increase at a constant rate r.
   2. Period 2: Consumes c_2, leaves bequest for future generations or not, and die.
2. Life-time utility: $U = u(c_1, l) + \delta \cdot v(c_2)$
3. Period budget constraints:

$$c_1 + s = wl \tag{1}$$

$$c_2 = s(1 + r) \tag{2}$$

4. Present value of lifetime income and consumption:

$$c_1 + \frac{c_2}{1 + r} = wl \tag{3}$$

Inter-temporal optimization

1. An individual works, earns and take all the productive decisions in period 1, and live off his savings in period 1 in the retirement period.
2. An individual's optimization problem:

$$\max_{c_1, l} \quad u(c_1, l) + \beta v(c_2) \text{ s.t } \quad c_1 + \frac{c_2}{1+r} = wl \tag{4}$$

3. Savings $s = wl - c_1$, $c_2 = (1+r)s$. Capital income rs
4. focs:

$$\text{Intra-temporal:labor supply:} \quad w\frac{\partial u}{\partial c_1} + \frac{\partial u}{\partial l} = 0$$

$$\text{Inter-temporal: savings} \quad \frac{\partial u}{\partial c_1} = \delta \cdot (1+r)\frac{\partial v}{\partial c_2}$$

Inter-temporal optimization conditions

- Intra-temporal condition of labour supply represents the equilibrium **within-period** trade-off between leisure and consumption. An unit of leisure leads to $\omega \frac{\partial u_1}{\partial c_1}$ amount of forgone first period consumption which should exactly offset each other at the equilibrium.
- Inter-temporal condition of **consumption smoothing** represents equilibrium across period trade-off between present and future consumption. An unit of consumption forgone in present gives an additional consumption of $(1 + r)$ in the next period and therefore $\delta(1 + r)\frac{\partial v}{\partial c_2}$ amount of utility in terms of today's consumption.

Concluding lecture
30th April, 2020

Taxes in life-cycle model

(a) Budget with consumption tax at rate t_c:

$$(1 + t_c)[c_1 + c_2/(1 + r)] \leq wl$$

(b) Budget with labor income tax at rate τ_L:

$$c_1 + c_2/(1 + r) \leq (1 - \tau_L)wl$$

(c) Consumption and labor income tax are equivalent if

$$1 + t_c = 1/(1 - \tau_L)$$

Both taxes distort only labor supply and not savings

taxes in life-cycle model

(d) With capital income tax at rate τ_K: $c_2 = (1 + r(1 - \tau_K)) \cdot s \Rightarrow$

$$c_1 + c_2/(1 + r(1 - \tau_K)) \leq wl$$

τ_K distorts only savings choice (and not labor supply)

(e) With comprehensive income tax τ on both labor and capital income: $c_1 = w(1 - \tau)l - s, \quad c_2 = (1 + r(1 - \tau))s$

$$c_1 + c_2/(1 + r(1 - \tau)) \leq (1 - \tau)wl$$

τ distorts both labor supply and savings
τ imposes "double" tax: on (1) earnings AND on (2) savings

Effects of capital tax on savings

Consider simpler model (fixed earnings w in period 1)

$$\max_{c_1, c_2} u(c_1) + \delta \cdot u(c_2) \quad \text{subject to} \quad c_1 + \frac{c_2}{1 + r(1 - \tau_K)} \leq w$$

Recall that $c_1 = w - s$ and $c_2 = [1 + r(1 - \tau_K)] \cdot s$ [draw graph]
Suppose τ_K increases and hence $1/[1 + r(1 - \tau_K)] \uparrow$
1) Substitution effect: price of $c_2 \uparrow \Rightarrow c_2 \downarrow$, $c_1 \uparrow \Rightarrow$ savings $s = w - c_1$ decrease
2) Income effect: consumer is poorer $\Rightarrow$ both c_1 and $c_2 \downarrow \Rightarrow$ savings s increase
Total net effect is theoretically ambiguous $\Rightarrow \tau_K$ has ambiguous effects on s

Optimal capital income taxation

Two broad types of models:
1) Life-cycle models: wealth is due solely to life-cycle savings
2) Models with bequests: wealth is due solely to inheritances

Optimal tax in life-cycle models

Government can use both a progressive labor income tax $T(wl)$ and a linear capital income tax τ_K
Individuals live 2 periods, earn in period 1, retired in period 2

$$\max_{c_1,c_2,l} u(c_1) - h(l) + \delta u(c_2) \quad \text{s.t.} \quad c_1 + \frac{c_2}{1 + r(1-\tau_K)} \leq wl - T(wl)$$

Individuals differ only according to their earning ability w
Government maximizes social welfare function based on individual utilities
Atkinson-Stiglitz JpubE'76 theorem: The optimal tax τ_K on capital income should be zero. Using a labor tax on earnings $T(wl)$ is sufficient.

Optimal tax in life-cycle model

Atkinson-Stiglitz' theorem shows that life-time savings should not be taxed, tax only labor income

Key intuition: in basic life-cycle model, inequality in life-time resources is due solely to differences in earnings ability. This inequality can be addressed with labor income taxation. Capital income taxation needlessly distorts saving behavior.

From justice view: seems fair to not discriminate against savers if labor earnings is the only source of inequality.

Inheritance: Estate taxation

Estate federal tax imposes a tax on estates above $11M exemption (less than .1% of deceased liable), tax rate is 40% above exemption (in 2018+)
Charitable and spousal giving are fully exempt from the tax
E.g.: if Bill Gates / Warren Buffet give all their wealth to charity, they won't pay estate tax
Popular support for estate tax is pretty weak ("death tax") but public does not know that estate tax affects only richest
Support for estate tax increase shots up from 17% to 53% when survey respondents are informed that only richest pay it (Kuziemko-Norton-Saez-Stantcheva AER'15 do an online Mturk survey experiment)

Taxation of inheritances: welfare effects

Inheritances (or gifts from living parents) raise difficult issues of social justice [see Kaplow 2001]:
(1) Inequality in inheritances contributes to economic inequality and individuals not responsible for inheritances they receive:
$\Rightarrow$ seems fair to redistribute from those who received inheritances to those who did not
(2) However, it seems unfair to tax the parents who worked hard (and already paid tax on income) to pass on wealth to children
Liberals emphasize (1) [taxing heirs] while conservative emphasize (2) [death tax]

Taxation of Inheritances: Behavioral Responses

Potential behavioral response effects of inheritance tax:
(1) reduces wealth accumulation of altruistic parents (and hence tax base) [no very good empirical evidence, Kopczuk-Slemrod 2001 suggest small effects]
(2) reduces labor supply of altruistic parents (less motivated to work if cannot pass wealth to kids) [no good evidence]
(3) induces inheritors to work more through income effects because they receive smaller inheritances (Carnegie effect, decent evidence from Holtz-Eakin,Joulfaian,Rosen QJE'93)
Critical to understand why there are inheritances for optimal inheritance tax policy. 3 models of bequests: (a) accidental, (b) altruistic bequests, (c) social/family pressure

Accidental bequests

People die with a stock of wealth they intended to spend on themselves (or that they accumulated out of love for wealth, Carroll '98):
Bequest taxation has no distortionary effect on behavior of parent and can only increase labor supply of inheritors (through income effects) $\Rightarrow$ strong case for taxing bequests heavily
Surveys show that bequest motives are not the main driver of wealth accumulation (Kopczuk-Lupton '07):
Only 1/3 of people surveyed say that the main reason they accumulate wealth is for bequests to their children

Altruistic bequests (Piketty and Saez 2013)

Utility $u(c) - h(l) + \delta v(b^{\text{left}})$ where c is own consumption, l is labor supply, and b^{left} is net-of-tax bequests left to next generation and $v(b^{\text{left}})$ is utility of leaving bequests for donor
Individual receives b^{received}, works and earns $wl - T(wl)$, consumes c, saves $s = wl - T(wl) + b^{\text{received}} - c$, which translates into $b^{\text{left}} = s(1+r)(1-\tau_B)$ for heir (τ_B is bequest tax rate)
Bequests provide an additional source of life-income:

$$c + \frac{b^{\text{left}}}{(1-\tau_B)(1+r)} = wl - T(wl) + b^{\text{received}}$$

In this model, Atkinson-Stiglitz breaks down and using bequest taxation is desirable to supplement labor income taxation
$\Rightarrow$ Two-dimensional inequality (labor,bequests) requires two-dimensional tax policy tool (labor tax, bequest tax)

Manipulative bequests

Parents use potential bequest to extract favors from children
Empirical Evidence: Bernheim-Shleifer-Summers JPE '85 show that number of visits of children to parents is correlated with bequeathable wealth but not annuitized wealth of parents
[Annuitized wealth is wealth that disappears at death such as a pension or an annuity]

$$\text{Visits}_i = \alpha + \beta \cdot \text{Bequeathable Wealth}_i + \gamma \cdot \text{Annuitized wealth}_i + \varepsilon_i$$

In regression, they find $\beta > 0$ and $\gamma = 0$ (but causality not clear)
$\Rightarrow$ Bequest becomes one additional form of labor income for inheritor and one consumption good for parent
$\Rightarrow$ Inheritances should be counted and taxed as labor income for donees

Social family-pressure bequests

Parents may not want to leave bequests but feel compelled to by pressure of heirs or society: bargaining between parents and children

With estate tax, parents do not feel like they need to give as much $\Rightarrow$ parents are made better-off by the estate tax $\Rightarrow$ Case for estate taxation stronger

Empirical evidence:

Aura JpubE'05: reform of private pension annuities in the US in 1984 requiring both spouses signatures when worker decides to get a single annuity or couple annuity: reform increases sharply couple annuities choice

Equal division of estates [Wilhelm AER'96, Light-McGarry '04]: estates are very often divided equally probably to avoid conflicts [gifts before death are not as equally split]